

HS 2022

Structural Design

D-BAUG - MIBS

Prof. Dr. Joseph Schwartz · Prof. Dr. Philippe Block
Federico Bertagna · Davide Tanadini · Dr. Ole Ohlbrock

Introduction



Prof. Dr. Joseph Schwartz



Prof. Dr. Philippe Block



Federico Bertagna



Davide Tanadini



Dr. Patrick Ole Ohlbrock

Teaching Team

Introduction

Week 1	22.9.	Global Equilibrium
Week 2	29.9.	
Week 3	06.10.	
Week 4	13.10.	Local Equilibrium
Week 5	20.10.	
Week 6	27.10.	
Week 7	03.11.	Design Exercise (midterm)
Week 8	10.11.	
Week 9	17.11.	Material and construction
Week 10	24.11.	
Week 11	01.12.	Design Exercise (final)
Week 12	08.12.	
Week 13	15.12.	
Week 14	22.12.	

Introduction

		Lecture	Exercise session / Task	Submission (Friday 23:59)
Week 1	22.9.	Introduction		
Week 2	29.9.		Basics of graphic statics	
Week 3	06.10.	Task 0		
Week 4	13.10.	Theory of Plasticity		Submission of Task 0
Week 5	20.10.		Strut-and-tie models	
Week 6	27.10.	Design exercise (Task 1)		
Week 7	03.11.	Design exercise (Task 1)		
Week 8	10.11.		Midterm review	Submission of Task 1
Week 9	17.11.	Stress fields		
Week 10	24.11.		From STM to stress fields	
Week 11	01.12.			
Week 12	08.12.	Design exercise (Task 2)		
Week 13	15.12.			Submission of Task 2
Week 14	22.12.	Final review		

Learning Goals

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4. Analyze and discuss a given structural design concept.

Learning Goals

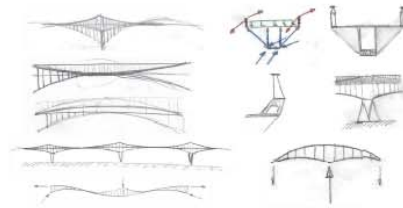
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4. Analyze and discuss a given structural design concept.
5. Propose transformations to improve a given design concept.

Learning Goals

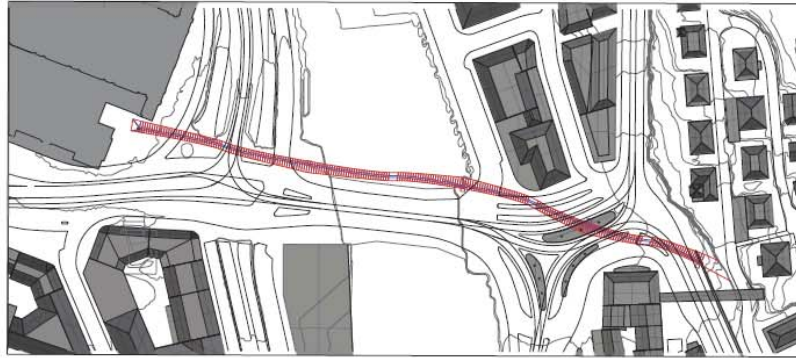
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3. Understand graphic statics and the relationship between the form of a structure and the internal forces within it.
4. Analyze and discuss a given structural design concept.
5. Propose transformations to improve a given design concept.
6. Generate appropriate design concepts and options for a given design task.

Introduction

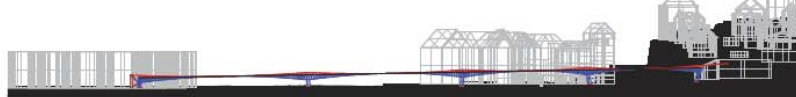
REBOUND

[illegible]

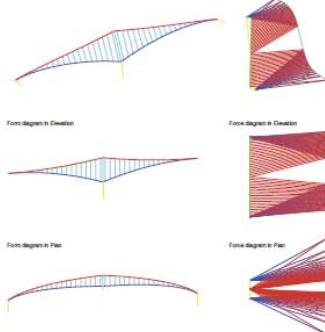
GROUND PLAN



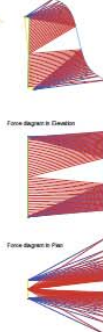
ELEVATION



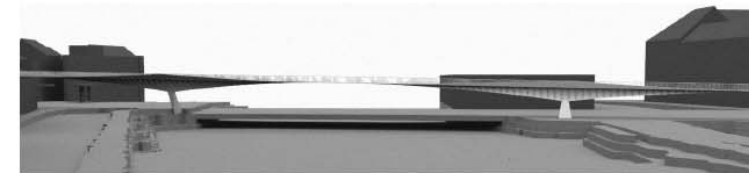
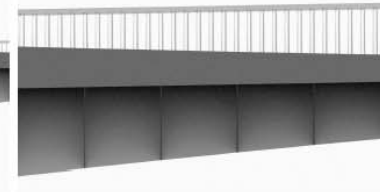
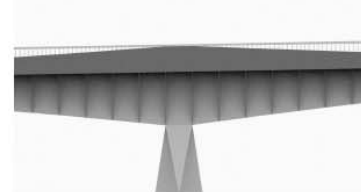
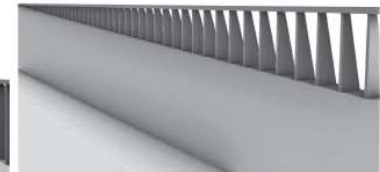
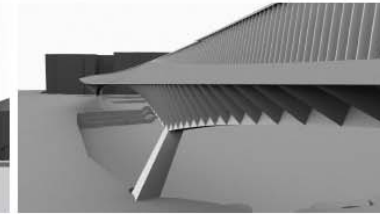
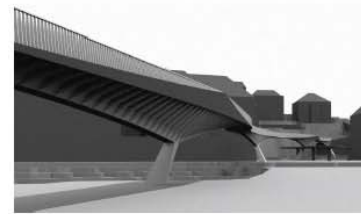
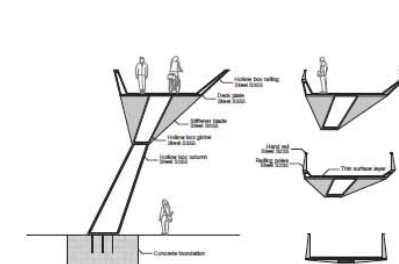
Forms designed in January



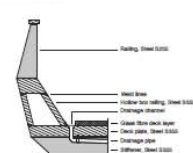
Force diagram is shown,



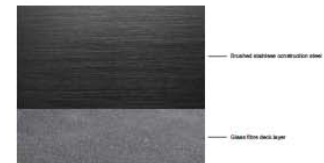
CROSS SECTION 1-YR



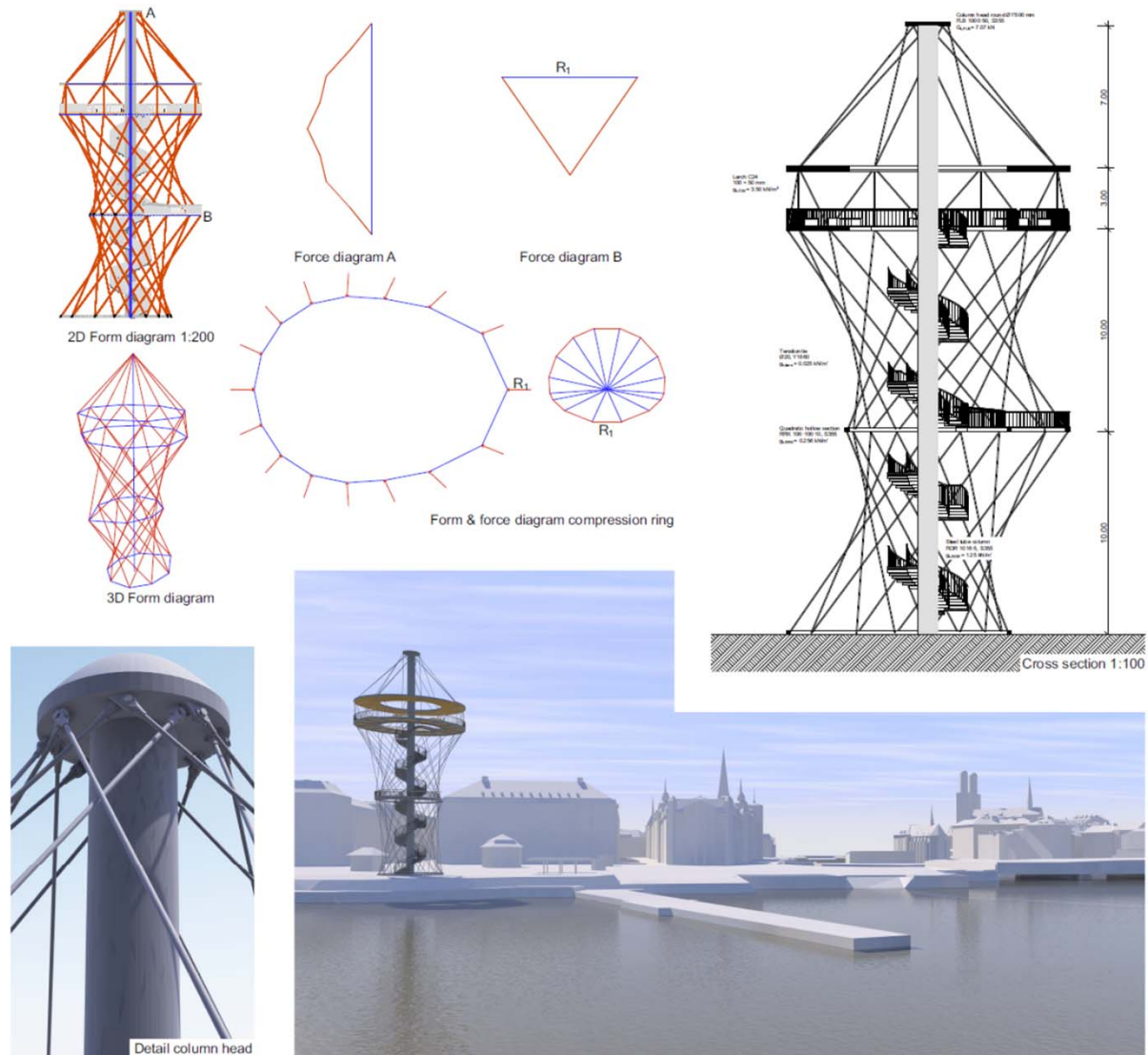
CONSTRUCTION DETAIL 120



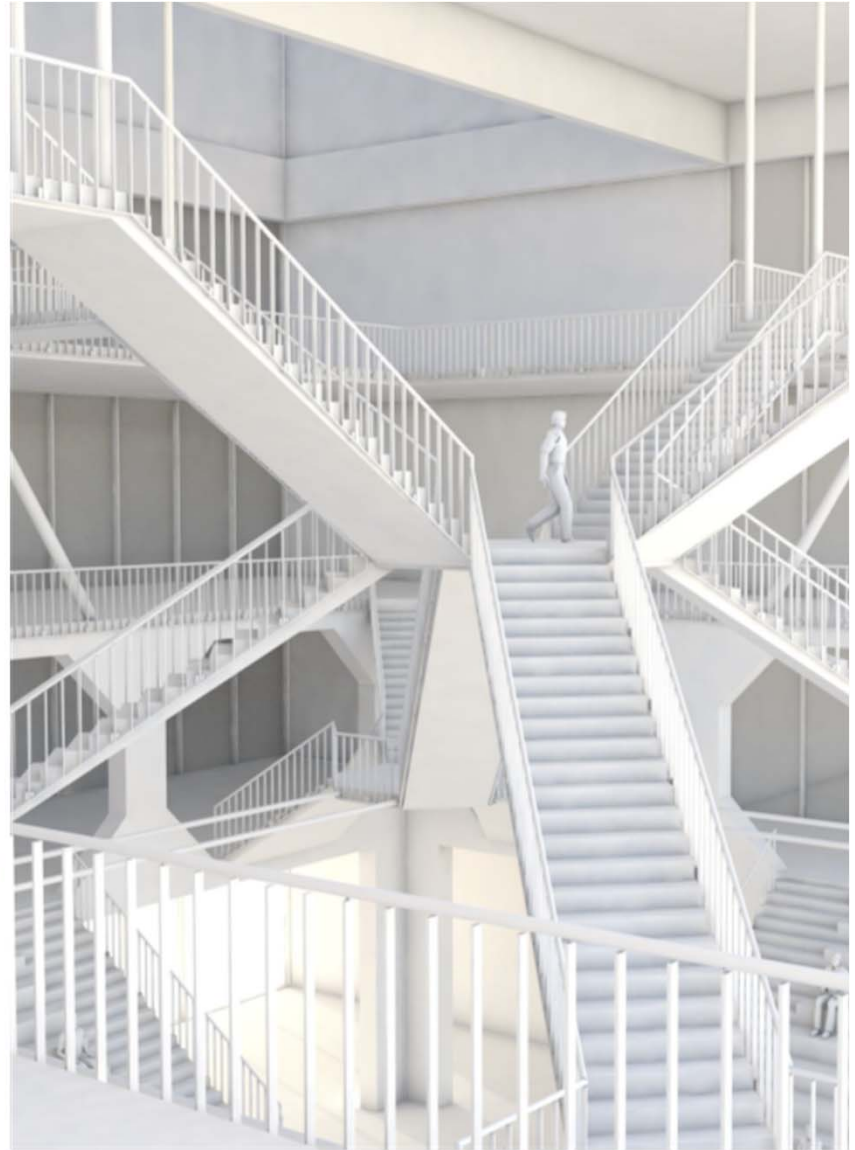
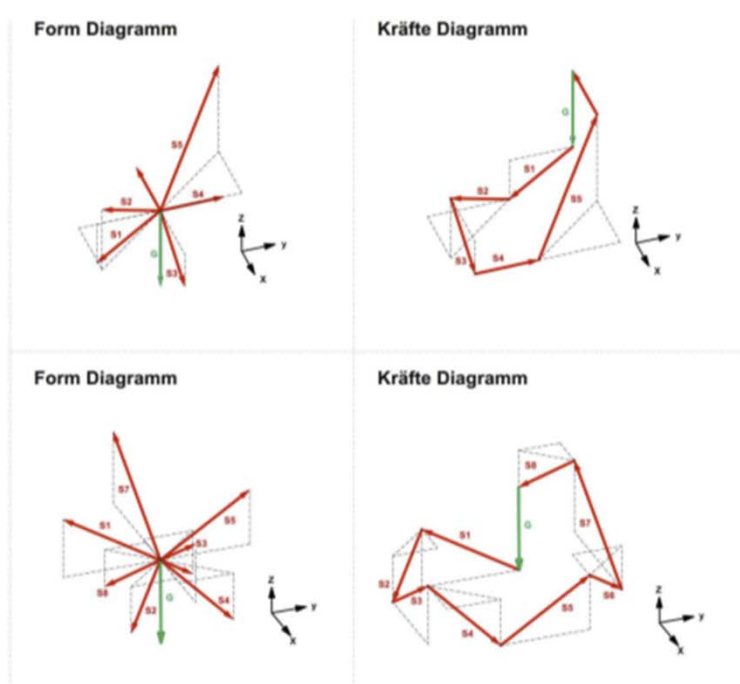
NEUTRALISATION



Introduction



Introduction



Introduction

Design exercise

Structural Design D-BAUG / MIBS - HS2021

Group: 2
Students: SILVIA TADDEI
LENA ALLGÖRIN

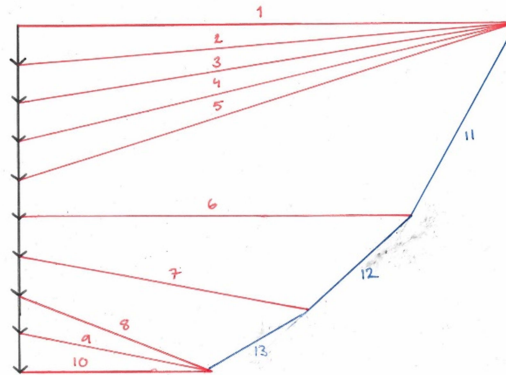
main structural elements and materials

DECK $\rightarrow$ combination of steel tie and stones in compression
PYLONS $\rightarrow$ glass and carbon fiber

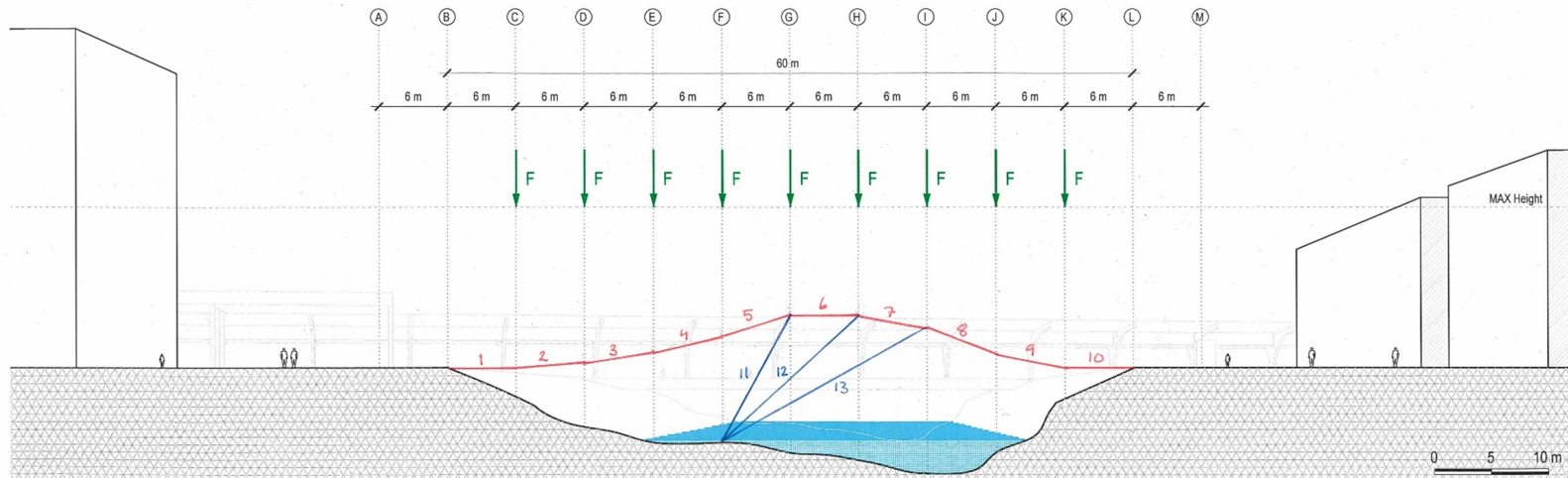
forces

edge	length [m]	force [kN]	edge	length [m]	force [kN]
1	6	12500			
2	5.8	12500			
3	5.3	13100			
4	5.6	13500			
5	5.3	13500			
6	6	14000			
7	5.3	7100			
8	5.6	5300			
9	5.3	5000			
10	6	4900			
11	12.6	5570			
12	16	3560			
13	20.3	3000			

total load path = 745.512 kNm



force diagram (scale 1: 1000)



Introduction



Site

Introduction



Physical model

 **eQUILIBRIUM**
an interactive environment for graphic statics-based structural design

my navigation ▼

examplescase studiesdrawings**courses**exams

Structural Design D-BAUG + MIBS

ENDE

Fall23.09.2021 → 23.12.2021

The goal of the course is to introduce students to Structural Design, which is regarded as a discipline that relates structural behavior, construction technologies and architectural concepts. The course encourages students to understand the relationship between the form of a structure and the forces within it by promoting the development of designed projects.

infothis weekscheduleslecturesexerciseprojects

<http://block.arch.ethz.ch/eq/course/182>